

1.5 Nested Quantifiers

[Example:] $\forall x \exists y (x+y=0)$. One quantifier is within the scope of another.

1.5.2 Understanding Statements Involving Nested Quantifiers

To understand statements involving nested quantifiers, we need to unravel what the quantifiers and predicates that appear mean.

[Example 1:] Assume that the domain for the variables x and y consists of all real numbers. The statement $\forall x \forall y (x+y=y+x)$ says that $x+y=y+x$ for all real numbers x and y . This is the commutative law for addition of real numbers.